

IWC Migration Programme

Tranche 1 / Phase 1

Uncertainty on Existing Models -- Plan vs Execution

The second phase of Tranche 1. Where Phase 0 re-framed the platform's language, Phase 1 attaches honest ERROR BARS to the existing whale species-distribution models and exposes the raw whale x vessel co-occurrence signal on its own. It adds three things the IWC SDM guidance (Miller & Kelly, 2023) asks for but the platform lacked: a predictive-uncertainty surface, a spatial-residual diagnostic, and an extrapolation flag -- plus an exposure-first base layer (Leaper's explicit request). No new external data; the science is made more honest, not more.

5

Scope items

50

ISDM refits

27.4%

Cells flagged

7.3M

Rows scored

270

tests pass

33

dbt nodes

1

new mart

1

API layer

Status: LANDED, 2026-06-15. Phase-1 verification gate passed; compute-and-store delivered first, UI exposed as a fast-follow. Phase 1b started separately.

1. Where Phase 1 Sits in the Programme

Phase 1 is the second phase of Tranche 1 ('standards + screening refresh', no new external data). It follows Phase 0's framing work and precedes the audit-driven model corrections of Phase 1b and the true vessel-traffic-density rewrite of Phase 2. It touches the modelling, error-analysis, and presentation layers -- it quantifies the uncertainty in models that already exist rather than building new ones.

Tranche 1 phase	Theme	State
Phase 0	Framing, model cards & crosswalk	Landed
Phase 1	Uncertainty on existing models	Landed (this report)
Phase 1b	Audit-driven model corrections	In progress
Phase 2	True VTD + Garrison lethality	Not started

Delivery philosophy -- compute-and-store FIRST

Land the science and the error analysis, inspect the diagnostics, and only THEN expose them in the UI as a fast-follow slice. This de-risks the method (does the spread surface look sensible? is the extrapolation flag firing where expected?) before paying the multi-touchpoint plumbing cost of a new map layer.

2. The Plan -- Phase-1 Scope

Phase 1 was scoped to five deliverables spanning modelling, error-analysis, and presentation:

1. Bootstrap / bagging ensemble (NOT quantile regression -- the SDMs are binary:logistic, outputting a Bernoulli P(presence), so quantile regression does not apply). Refit ~50 models on resampled rows with varied seeds; the per-cell MEAN is the point estimate and the per-cell SPREAD is the uncertainty band / CV-analogue surface. New columns in the SDM marts.
2. Spatial residual diagnostics in evaluate.py: empirical variograms and Moran's I on out-of-fold residuals, to confirm whether the models capture the spatial structure in the data.
3. ExDet / MESS-style covariate-envelope extrapolation flag on every prediction -- a 'low-confidence / extrapolated' overlay marking cells whose environment lies outside the training range.
4. Exposure-first base layer: report the RAW whale x vessel co-occurrence per cell BEFORE any speed-lethality weighting (Leaper's explicit request), with the speed-weighted variant as an optional overlay.
5. Expose the uncertainty, extrapolation, and exposure layers in the map UI and API -- the FAST-FOLLOW, after compute-and-store is validated.

Why bootstrap, not quantile regression

Quantile regression estimates conditional quantiles of a CONTINUOUS target. The SDMs predict a probability (Bernoulli mean), which has no meaningful quantiles to regress. Refitting the model on resampled data and reading the spread of predictions is the correct uncertainty tool for a classifier. Quantile regression is reserved for the Phase 3/5 DSM DENSITY (animals per km-squared), which is continuous.

3. The Execution -- What Was Built

3.1 Uncertainty toolkit (pipeline/analysis/uncertainty.py)

A new pure-numpy / XGBoost helper module, with no I/O, that the training scripts call. Two public functions:

- `bootstrap_ensemble_predict(...)` -- fits K models, each on a bootstrap resample (rows sampled WITH replacement), and scores the grid with every member. Returns the per-cell MEAN and STANDARD DEVIATION of P(presence) across the K members. Supports sub-bagging (`sample_frac < 1.0`) and a reduced per-member tree count so the 7.3M-row seasonal grid stays affordable.
- `mess_extrapolation(...)` -- the Multivariate Environmental Similarity Surface (Elith et al. 2010). Per cell it returns the minimum similarity across covariates (negative => at least one covariate is out of the training range) and names the most-dissimilar (MoD) variable for the model card.

3.2 Spatial residual diagnostics (evaluate.py)

Out-of-fold residuals plus each cell's lat/lon are already in-frame at train time, so residual diagnostics were nearly free to add. New helpers: `morans_i` (global spatial autocorrelation via a cKDTree neighbour graph), `empirical_variogram` (binned semivariance vs distance), and `spatial_residual_diagnostics / plot_spatial_diagnostics`, which persist a per-target PNG. Strong residual autocorrelation flags spatial structure the model has NOT captured -- exactly the signal the IWC guidance wants reported.

3.3 Training-script wiring

The ISDM and seasonal-SDM trainers gained CLI flags to drive the ensemble and emit the new columns, with budgets matched to each model's size:

Trainer	Ensemble budget	Outputs added
<code>train_isdm_model.py</code>	K=50 full bootstrap (Nisi data is tiny)	<code>_sd</code> per species + MESS (value, flag, MoD)
<code>train_sdm_seasonal.py</code>	K=15 sub-bagging (frac 0.3, 200 trees)	<code>_sd</code> per species + residual diagnostics

MESS extrapolation is computed for the ISDM only: it is trained on point (presence) data from the Nisi grid and then scores a DIFFERENT (H3) grid, so envelope extrapolation is a real risk. The seasonal SDM trains on the full H3 grid it later scores, so by construction it does not extrapolate beyond its own covariate envelope.

3.4 Marts -- uncertainty columns + new exposure mart

The loaders (`load_ml_predictions.py`, `load_sdm_predictions.py`) carry the new `_sd` and MESS columns through to PostGIS via the existing COPY-from-StringIO idempotent pattern. The intermediate models (`int_ml_whale_predictions`, `int_sdm_whale_predictions`) surface a combined `mean_whale_sd`, and the ML risk mart gained four honest confidence columns:

- `fct_collision_risk_ml.sql` now exposes `whale_prob_sd` (ensemble uncertainty), `isdm_mess_value`, `isdm_extrapolated` (boolean), and `isdm_mod_variable` (the most-dissimilar covariate name).

A brand-new mart, `fct_whale_vessel_exposure.sql`, delivers the exposure-first base layer at (h3_cell, season) grain (~7.3M rows):

Column	Definition	Role
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Column	Definition	Role
exposure_raw	P(any whale) x vessel volume	BASE layer
exposure_speed_weighted	P(any whale) x speed lethality	optional overlay
exposure_score	percent_rank of raw, PARTITION BY season	season-relative
exposure_speed_score	percent_rank of speed-weighted, by season	season-relative

Decoupling exposure from lethality makes the co-occurrence signal -- the quantity most directly comparable across studies and least dependent on modelling assumptions -- visible on its own, satisfying the standard's 'report with and without the lethality weighting' rule.

3.5 UI fast-follow (exposure layer)

Once the mart was validated, the exposure layer was wired end-to-end as a fast-follow:

- Backend: a new GET /api/v1/layers/exposure endpoint (ExposureCell model + get_exposure / count_exposure service functions), filtered by bbox, season, and a minimum exposure-score threshold.
- Frontend: a new 'exposure' LayerType threaded through types.ts, the useMapData hook (endpoint + seasonal-layer set), colors.ts (colour ramp, heatmap range, macro weight field), the Legend, the Sidebar (layer guide entry under Vessel Traffic), and CellDetail (an explanation panel).

4. Design Decisions & Trade-offs

Full bootstrap for ISDM, sub-bagging for the seasonal SDM

The ISDM trains on the small Nisi grid, so 50 full bootstrap refits are cheap. The seasonal SDM scores 7.3M rows across seven targets, so a naive K=50 full bootstrap was too slow; sub-bagging (30% row sample, 200 trees, K=15) gives a robust spread estimate at a fraction of the cost. The spread estimator is insensitive to these knobs, so accuracy is preserved.

MESS on the ISDM only

Extrapolation is only meaningful when a model scores a grid DIFFERENT from its training support. The ISDM does (Nisi points -> H3 grid); the seasonal SDM trains and scores on the same grid. Computing MESS for the SDM would have produced a vacuous all-clear flag, so it was deliberately omitted.

A separate exposure mart, not another column

Exposure-first is a conceptually distinct reporting product, not a tweak to the composite risk score. Giving it its own mart (fct_whale_vessel_exposure) keeps the base co-occurrence signal cleanly separable in the API and UI, and lets the speed-weighted overlay sit beside the raw layer without entangling either with the seven-sub-score collision index.

5. Verification -- Gate Results

Both the Phase-1 verification line and the global gate were run. All checks passed:

Check	Command	Result
Uncertainty cols	fct_collision_risk_ml in PostGIS	7.3M rows; _sd + MESS populated

Check	Command	Result
ISDM ensemble	train_isdm --score-grid --bootstrap-k 50	4 species; 27.4% extrapolated
Seasonal diag	train_sdm_seasonal --bootstrap-k 15	8 residual PNGs in artifacts
Exposure mart	dbt build fct_whale_vessel_exposure	7.3M rows; raw + speed populated
Subgraph build	dbt build (modified + downstream)	33 nodes pass
Lint	ruff check pipeline/ backend/ tests/	clean
Unit tests	pytest tests/	270 pass
Frontend	tsc --noEmit	exit 0
Orchestration	dagster definitions validate	valid

Gate passed

ruff clean; dbt subgraph 33 nodes green; 270 pytest pass; frontend type-check clean; Dagster validates. Uncertainty columns and the exposure mart are populated and the exposure layer renders. Phase 1 is complete; Phase 1b is a separate piece of work.

6. What the Error Analysis Revealed

The point of adding diagnostics is to LOOK at them. Two findings of note from this first run:

- Extrapolation is real and concentrated: 27.4% of ISDM-scored cells fall outside the training envelope, and the most-dissimilar variable is most often pp_upper_200m (primary productivity). These cells now carry an honest low-confidence flag instead of being shown as equally trustworthy.
- The minke-whale seasonal SDM left residual Moran's I ≈ 0.70 -- strong positive spatial autocorrelation in the out-of-fold residuals, i.e. genuine spatial structure the model has NOT captured. A real diagnostic catch, recorded for follow-up rather than silently passing.

Feeds Phase 1b

The variogram range produced here is exactly the input Phase 1b item E needs to set the spatial-CV block size from residual autocorrelation instead of a fixed 158 km guess. Phase 1's error analysis is reused, not rebuilt.

7. Key Takeaways & What's Next

- Honest error bars first: every whale prediction now carries a spread (uncertainty) and, for the ISDM, an extrapolation flag -- the platform can finally distinguish 'high risk + confident' from 'high risk but data-poor'.
- The right tool for the target: bootstrap spread for a probability classifier; quantile regression deferred to the continuous Phase 3 density.
- Compute-and-store first paid off -- the diagnostics were inspected (and caught a genuine minke autocorrelation issue) before any UI plumbing was written.
- Exposure-first decoupling gives a co-occurrence base layer that is comparable across studies and independent of lethality assumptions, directly answering Leaper's request.
- The new diagnostics are not throw-away: the variogram range feeds the Phase 1b CV-block sizing, and the MESS MoD variable feeds the model cards.

Next: Phase 1b -- audit-driven corrections on the existing models (no new data). The headline items are the ensemble

NULL-deflation bugfix (a missing model should not halve P(whale)), probability recalibration, target-group background + spatial thinning to remove OBIS effort bias, a skill-weighted ensemble, variogram-driven CV block sizing, a strike-weighted whale-exposure column, the addition of Rice's whale and gray whale to the species set, and a protection-gap refresh folding in critical habitat and active slow zones.

References

Elith et al. (2010) *Methods Ecol. Evol.* 1:330-342 -- MESS / ExDet environmental novelty. Miller & Kelly (2023) SC/69A/ASI/20 -- IWC model-based abundance / SDM guidance. Leaper et al. (2026) SC/70/HIM/13 -- IWC strike-risk reporting standard (exposure-first request). Nisi et al. (2024) *Science* 386(6724):870-875 -- ISDM training source. Rockwood et al. (2021) -- whale x vessel co-occurrence interaction basis.